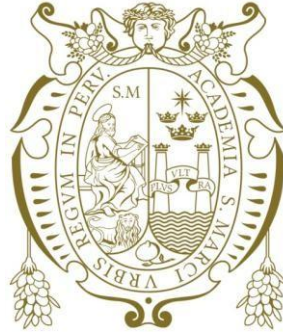


UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Doctoral Program in Deep Tech, Artificial



Intelligence & Emerging Technologies

**Research Methods and Scientific Integrity in AI and
Advanced Technologies**

METHOD FIT MATRIX DELIVERABLE No. 2

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Scoped research question: How can a verifiable, human-in-the-loop generative AI artefact be designed and rigorously evaluated to recommend contractual clauses whose every output is traceable to a contrastable normative or contractual source in one SBS-supervised fiduciary institution and one clearly bounded clause-recommendation task and does it outperform a standard retrieval-augmented generation (RAG) baseline on verifiability and hallucination control?

Criterion / Weight	Pure Experiment	Pure Case Study	Mixed Methods	Design Science Research
E: Epistemological Fit 30%	3/5 — Can test whether the artefact beats a RAG baseline, but does not govern how the verifiable artefact is designed.	2/5 — Strong on institutional and regulatory context, but weak fit with the artefact-centred knowledge claim.	4/5 — Integrates benchmark and expert evidence, but is broader than the core prescriptive contribution.	5/5 — Natural fit: knowledge is generated by building and evaluating the artefact; verifiability translates refutability to the technical plane.
D: Data Feasibility 20%	2/5 — Experiment-scale, de-identified contract data with legal gold-standard labels are hard to obtain under banking secrecy.	3/5 — Deep access to a single institution is possible but resource-intensive.	3/5 — Requires both the quantitative benchmark and qualitative expert (lawyer) recruitment.	4/5 — Prototype logs, retrieval traces, attribution annotations and expert spot-checks are obtainable.
F: Feasibility 15%	2/5 — High confidentiality, annotation and control burden for an adequately powered experiment within three years.	3/5 — Feasible, but prolonged immersion and permissions add time.	3/5 — Feasible only if tightly scoped; a full mixed-methods study risks overextension.	4/5 — Feasible if limited to one prototype, one institution, one clause task and 2–3 outcomes.
C: Contribution 20%	3/5 — Produces effect evidence but not the intended design knowledge.	2/5 — Explains context but does not deliver the intended technological artefact.	4/5 — Produces integrated evidence, but artefact and design knowledge are not central.	5/5 — Delivers the artefact plus rigorous evaluation plus transferable design principles for verifiable generation in regulated domains.
V: Venue Fit 15%	4/5 — Publishable in NLP and evaluation venues if the benchmark design is defensible.	3/5 — Possible in IS and legal-tech venues, but a weaker match to the technical contribution.	4/5 — Well accepted in IS and AI venues, but with a broader study burden.	5/5 — Strong fit with MISQ, JAIS, EJIS and DESRIST, and with Artificial Intelligence and Law (ICAIL/JURIX).
Weighted Total Ex.30 + Dx.20 + Fx.15 + Cx.20 + Vx.15	2.80/5 (56/100)	2.50/5 (50/100)	3.65/5 (73/100)	4.65/5 (93/100)

Decision: Design Science Research is selected as the primary method. Pure Experiment and Pure Case Study are eliminated by the ≤ 2 gate. Mixed Methods remains a viable evaluation strategy embedded inside the DSR cycle, but not the governing method. To remain feasible within three years, the DSR study is restricted to one functional prototype, one SBS-supervised institution, one bounded clause-recommendation task, and 2–3 primary outcomes (verifiability/attribution quality, hallucination rate, and recommendation quality). A narrowly scoped controlled comparison against a standard RAG baseline with no-retrieval and current-practice baselines and focused expert feedback are embedded inside the DSR evaluation cycle.

Gate rule: any candidate scoring ≤ 2 on a critical criterion is removed. Simulation / synthetic-data study: not shortlisted as the primary method because the central claim concerns verifiability against real normative sources and real institutional auditability; it may still be used for pre-deployment stress testing of hallucination and mis-attribution failure modes.

Colour legend: Red indicates weak methodological fit (≤ 2); amber, moderate fit (3); green, strong methodological fit (≥ 4); blue, structural (weighted totals and criteria).

Data sources for the selected DSR plan: prototype interaction logs; retrieval traces (retrieved passages per query); generated clauses with their citations; citation precision and recall labels; span-level attributability judgments; hallucination annotations (unsupported, misattributed, or contradictory); the annotated legal gold standard; and focused lawyer/specialist feedback. Institutional authorization, banking-secrecy compliance (Ley 26702), data protection (Ley 29733), de-identification of contract text, local/on-premise processing where feasible, and human authority over any adopted clause are prerequisites.